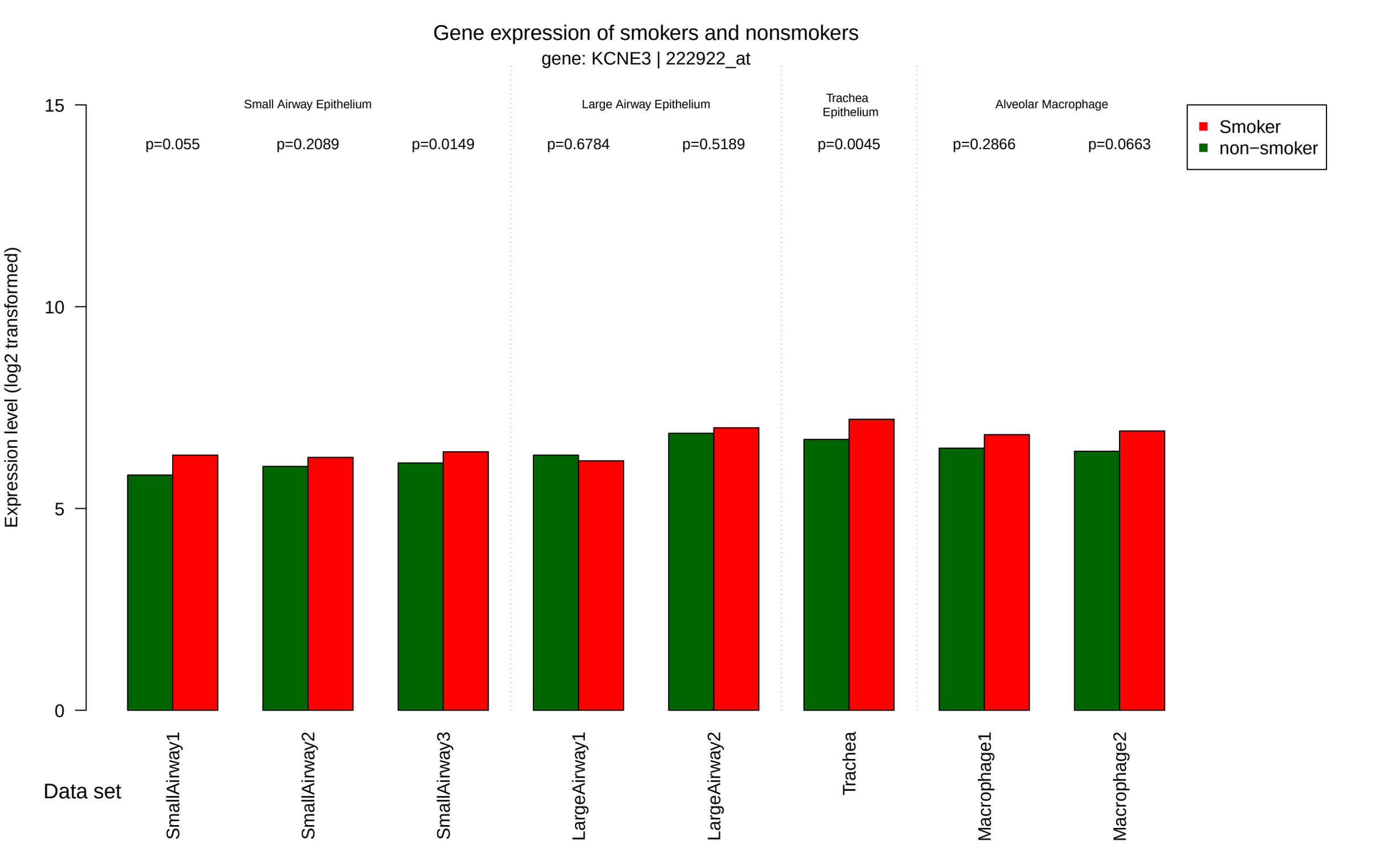
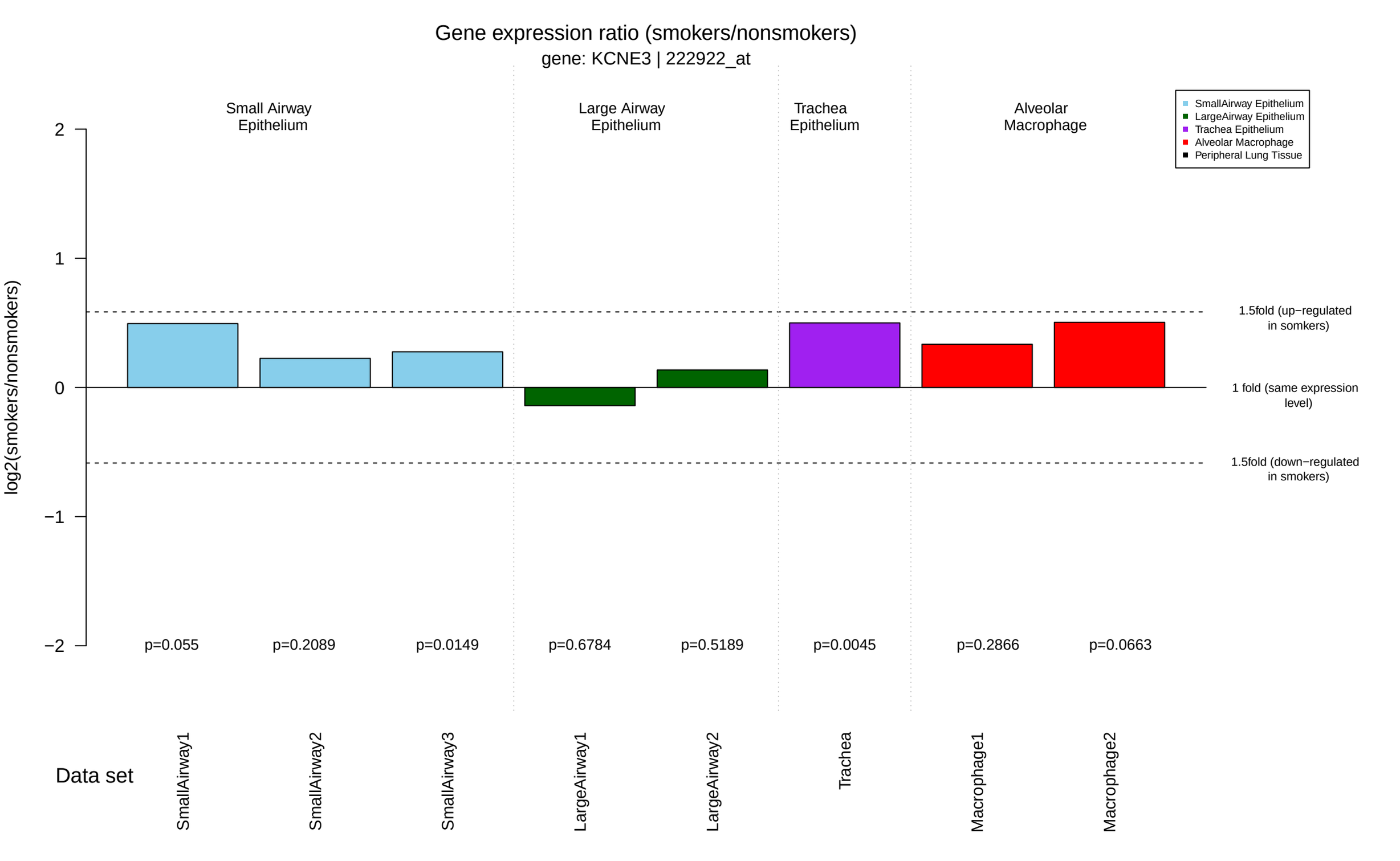
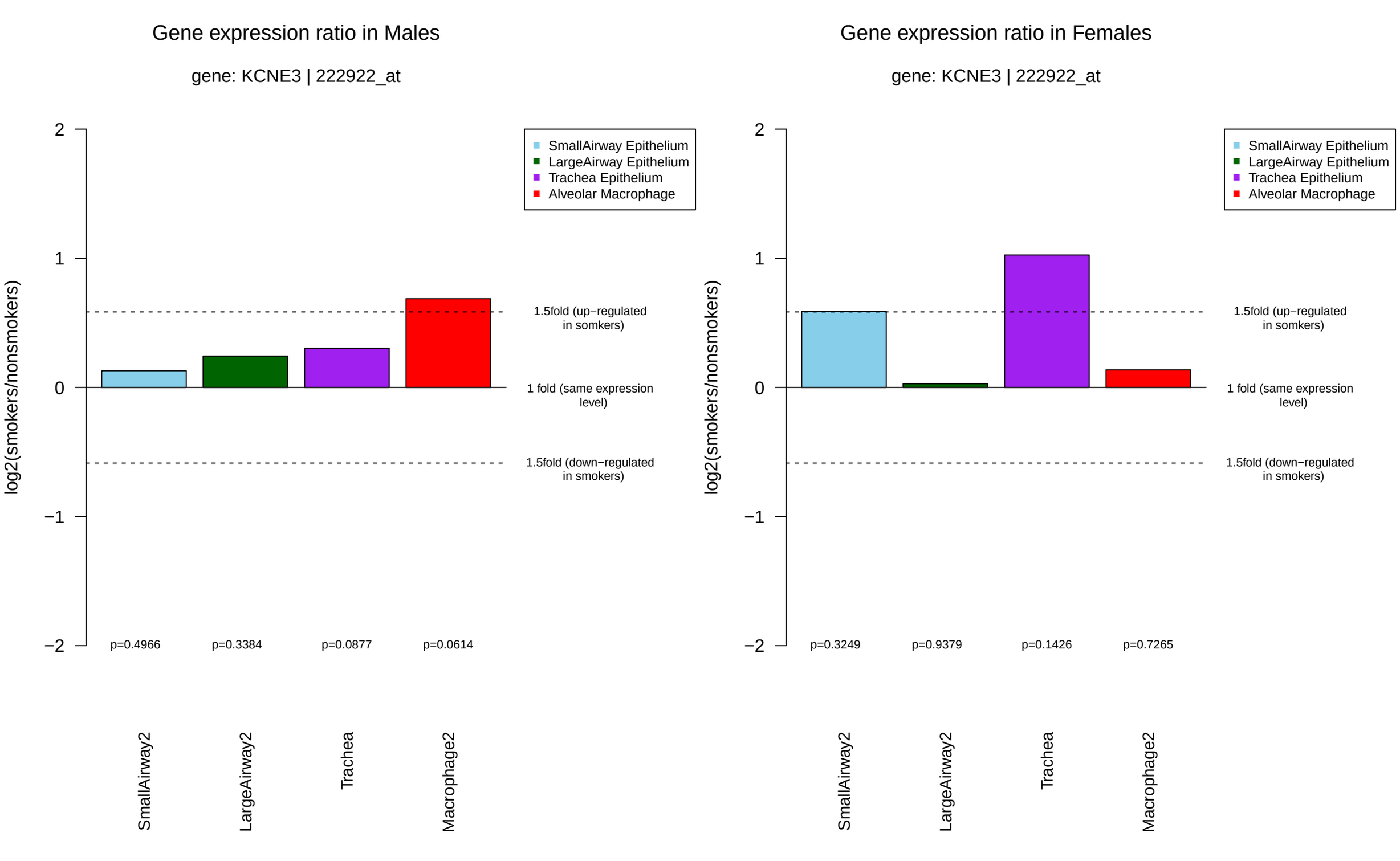
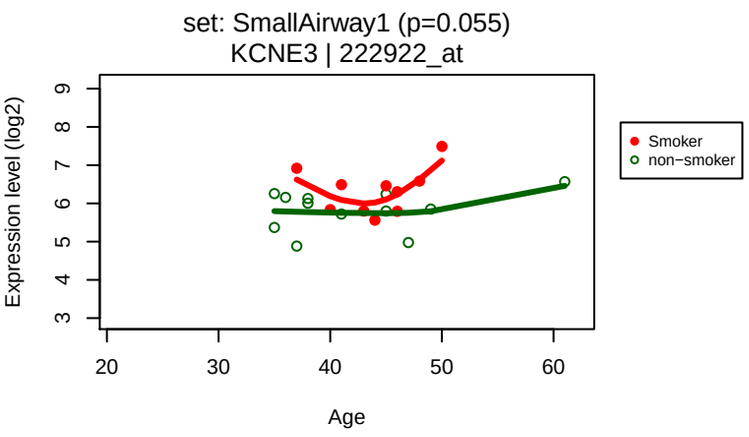
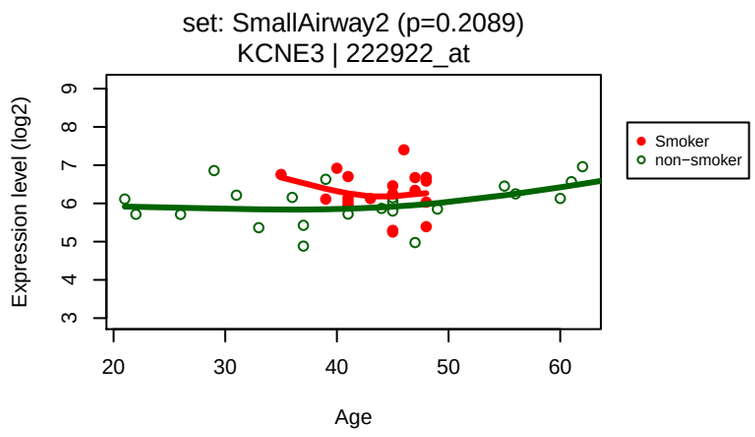
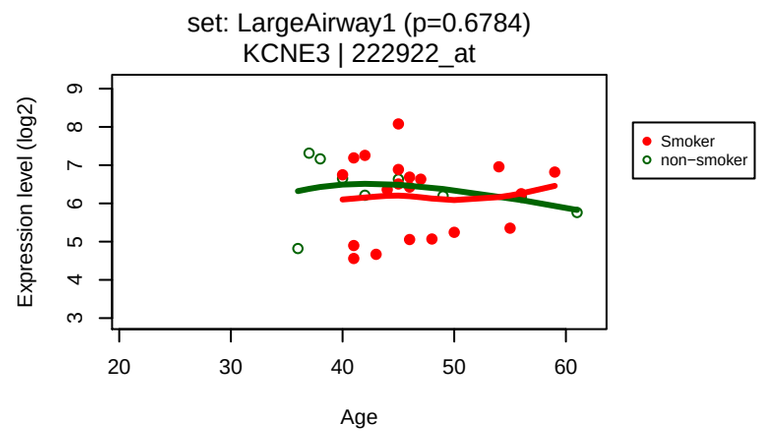

Gene expression of smokers and nonsmokers
gene: KCNE3 | 222922_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.055
p=0.2089
p=0.0149
p=0.6784
p=0.5189
p=0.0045
p=0.2866
p=0.0663
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

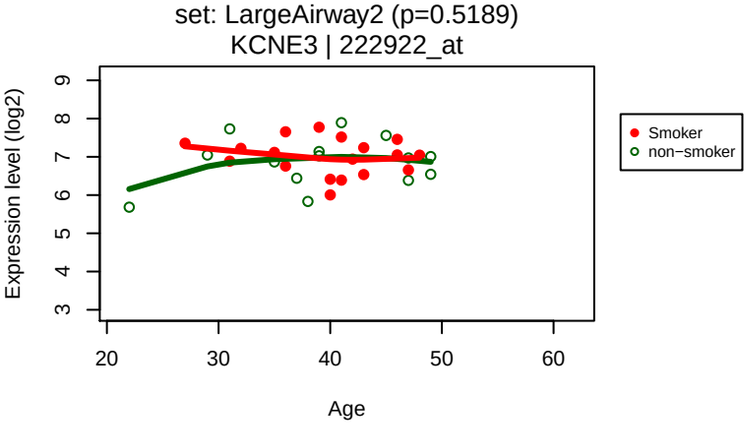

Gene expression ratio (smokers/nonsmokers)
gene: KCNE3 | 222922_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.055
p=0.2089
p=0.0149
p=0.6784
p=0.5189
p=0.0045
p=0.2866
p=0.0663
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

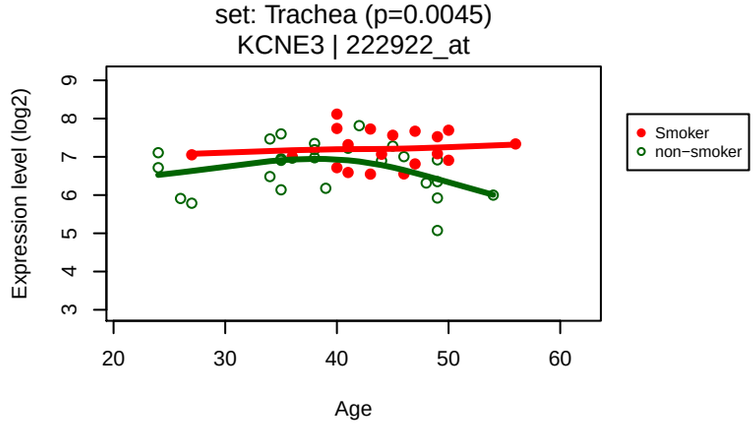

Gene expression ratio in Males
gene: KCNE3 | 222922_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4966
p=0.3384
p=0.0877
p=0.0614
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KCNE3 | 222922_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3249
p=0.9379
p=0.1426
p=0.7265
SmallAirway2
LargeAirway2
Trachea
Macrophage2

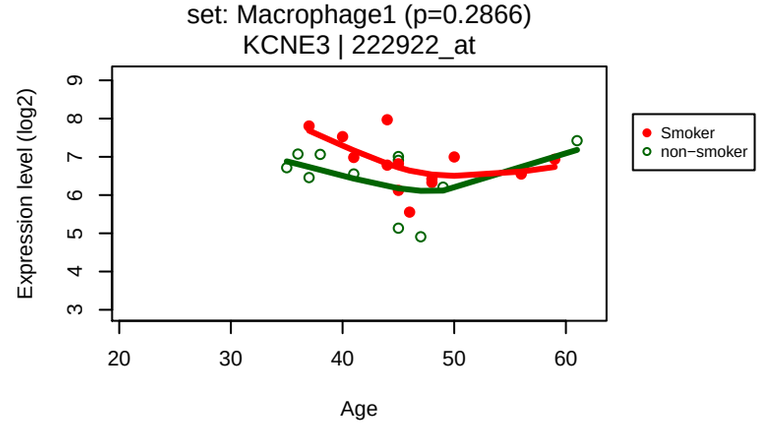

set: SmallAirway1 (p=0.055)
KCNE3 | 222922_at
Expression level (log2)
Age
Smoker
non−smoker

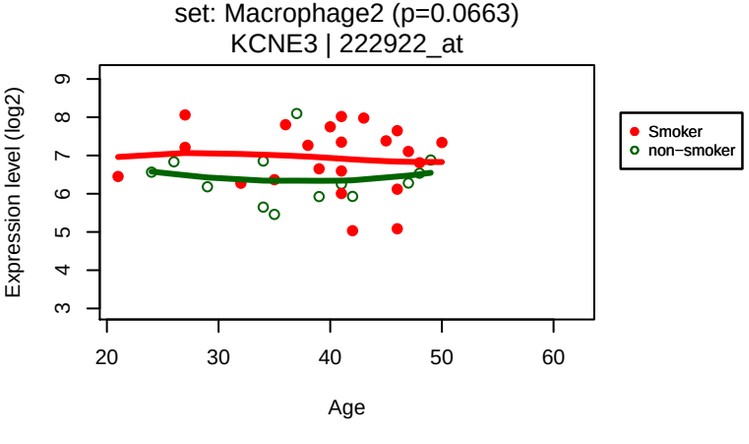

set: SmallAirway2 (p=0.2089)
KCNE3 | 222922_at
Expression level (log2)
Age
Smoker
non−smoker

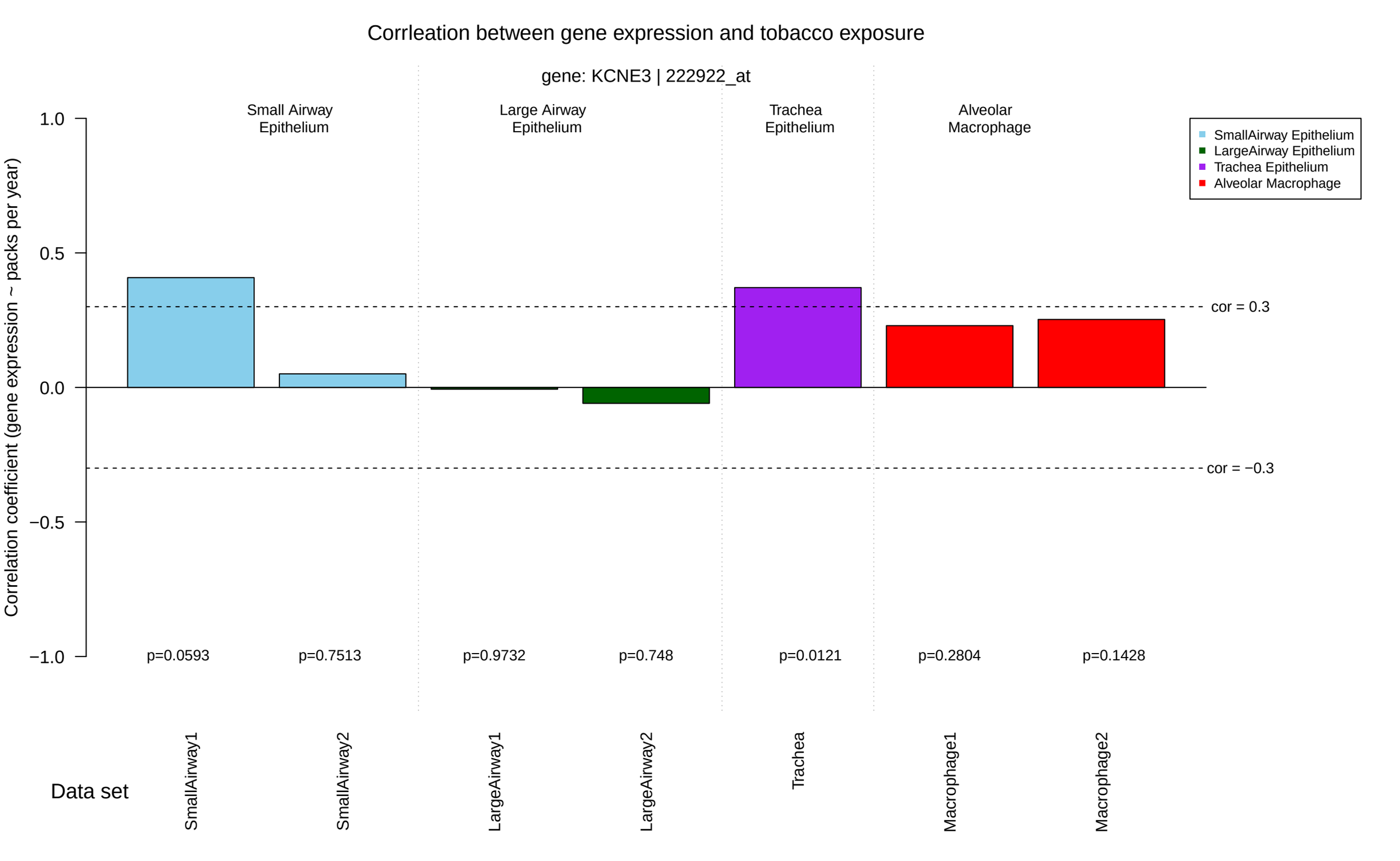

set: LargeAirway1 (p=0.6784)
KCNE3 | 222922_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5189)
KCNE3 | 222922_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0045)
KCNE3 | 222922_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2866)
KCNE3 | 222922_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0663)
KCNE3 | 222922_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KCNE3 | 222922_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0593
p=0.7513
p=0.9732
p=0.748
p=0.0121
p=0.2804
p=0.1428
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2